

Features Extraction Technique Analysis of EEG Signal for Motor Imagery Classification

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Abstract—For the daily life application of BCI, any assumptions could be violated during the practical test, especially with the small number of channels. Multiple electroencephalogram (EEG) channels could have high accuracy but are not practical for daily life usage compared to a single-channel EEG. However, the single-channel EEG signal has limited sources to extract informative features. Therefore, the aim of the study is to analyze the significant impact of different feature extraction techniques on single EEG channel classification performance. The BCI competition III (IVa) dataset is used for this purpose. Time-frequency domain (E1), and time-domain feature extraction techniques (E2) as well as the combination of both techniques (E3) have been considered to extract the relevant features set from the EEG signals for MI tasks classification. The classification accuracy is obtained with two classifiers: Support Vector Machine (SVM) and Logistic Regression (LR). The result shows that there is an enhancement in classification accuracy on certain channels and participants, with E3 being higher than E1 and E2. The C3 channel in E3 has the highest accuracy (87.1%) when SVM is the classifier. The proposed approach improves the classification accuracy of AF4 by up to 70% and 70.4% by using SVM and LR, respectively, which is considered acceptable for usage in a BCI system. This implies that MI tasks can be detected with a single-channel EEG by using the proposed method even if the channel is located on the forehead.

Keywords—Motor Imagery, Single-Channel EEG, time-domain features, time-frequency features, BCI

I. INTRODUCTION

Using a brain-computer interface (BCI), brain signals are collected and converted into machine commands that reflect the user's intention [1]. It is beneficial to have a device to be controlled by only her thoughts as an alternative to assisting patients with severe motor impairments [2], [3]. A user's perception, consciousness, and cognition are investigated through passive BCIs such as drowsiness detection [4]. An

active BCI generates patterns from brain activity that the user may directly and consciously control such as motor imagery (MI) [5]. To operate the MI-based BCIs, the user imagines the movement without performing it. Sensorimotor rhythm (SMR) patterns change because of movement or imagined movement. The components of SMR are μ rhythm (8–13 Hz) and β rhythm (13–30 Hz) [6], [7]. Left, right, forward, up, and down were the common movements or commands employed in MI tasks for wheelchairs and cursors [8].

Most BCI studies have used multi-channel electroencephalogram (EEG) data to achieve highly accurate results. With certain limits on MI tasks, the MI-BCI-based analysis produced good control on actual devices, including 3-D. However, there are still substantial limitations when only a few EEG channels were used [9]. The multi-channel analysis method, on the other hand, have increased the complexity of the experimental procedure. [10] employed two public datasets in their study to reduce the number of channels. The EEG channels that were substantially associated with the subject were identified using a Pearson correlation coefficient. When C3 was used as a reference channel, they discovered that subject A05 (BCI Competition III dataset IVa) performed well. Even though there were just 13 channels instead of 118, it was reported that the classification accuracy was still above 70% (80.56%). The number of channels required for a high accuracy rate becomes subject-specific, that is, it is highly influenced by the subject themselves as well as the good design of experimental conditions and classification algorithm [11]. The development of portable systems with fewer channels but reliable for signal acquisition, and ease of maintenance is essential in the BCI technology [12]. Therefore, it will be simpler and more comfortable to apply BCIs for daily use if high-accuracy classification can be obtained with just one EEG channel, particularly in the forehead [13]. However, the limitation of a single channel is to extract relevant features when compared to a multichannel that incorporates features from multiple channels.

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Therefore, the aim of this study is to analyse the single-channel EEG signal classification accuracy with different feature extraction techniques. To achieve the aim, the EEG signal from the channels on the forehead (Fp1, Fp2, AF3, and AF4) is selected. They are considered practical for daily life usage. Other than that, the signals from the commonly used channels (C3 and C4) in MI, are also chosen. The classification performance with the set of time-frequency domain features, time-domain features, and a combination of both features are investigated for any possibility of accuracy enhancement. Since Wavelet Transform (WT) demonstrated greater robustness than Common Spatial Pattern (CSP) and Power Spectral Density (PSD) [14], Discrete Wavelet Transform (DWT) is employed for the time-frequency features set.

II. RELATED WORKS

Time-frequency decomposition techniques are one approach for obtaining more time-varying signals at various frequency bands from the single-channel EEG signal. Short-time Fourier transforms (STFT) and wavelet transform (WT) are commonly used for time-frequency transform. STFT was applied in previous studies of a single-channel analysis [13], [15], [16]. It allows the implementation of the CSP method that needs a greater number of channels [13]. Novel frameworks based on an enhanced two-dimensional nonlinear FitzHugh-Nagumo (FHN) were proposed for extracting relevant features from the single-channel MI. By using STFT, FHN-PSD features were log-scaled variations of PSD, whereas FHN-STFTCSP features were log-scaled variations of CSP [17]. They found that the methods obtained higher accuracy with Support Vector Machine (SVM) as the classifier. However, STFT is not suitable for non-stationary because of the fixed window size [18].

DWT decomposes the signal into multiple bands of frequency. It is useful for non-stationary signals. It has been combined with multiple methods to extract relevant features to improve the accuracy of classification algorithms used to differentiate between various types of motor imagery tasks. An approach based on DWT and Empirical Mode Decomposition (EMD) along with approximative entropy for feature extraction was proposed by [19]. The features from the channels of C3 and C4 were the input to the SVM to carry out the task's classification. The proposed method had a limitation in that the statistic for approximate entropy was inconsistent. In other applications, a combination of two sets of features was applied by concatenating DWT and EMD-based features matrix for seizure detection over multi and single-channel EEG signals [20]. It produced promising results, but this method could be time-consuming because it uses a lot of computational power to process all the features obtained from the two decomposition techniques.

III. METHODOLOGY

The block diagram of the proposed approach is shown in Fig.1. At the feature extraction level, it consists of three experiments named E1, E2, and E3. All levels are described in detail in the following subsections:

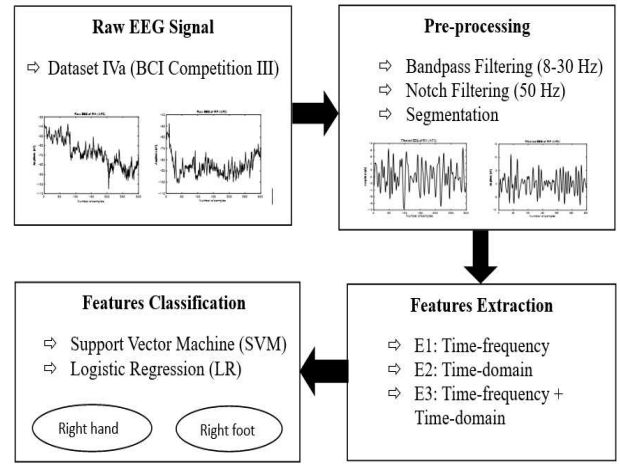


Fig. 1. Block diagram of the proposed approach.

A. Dataset

This approach employed the cued motor imagery-related data collection IVa from the third BCI Competition [21]. 118 EEG channels from the extended 10-20 international system were recorded. The data set included EEG signals from motor imagery tasks performed while moving the right hand (*rh*) and right foot (*rf*). The signals were acquired from five healthy participants (*aa*, *al*, *av*, *aw*, and *ay*). A visual stimulus was displayed for 3.5 seconds while the participant comfortably sat during the experiment. Each participant completed 280 trials with 140 trials for each task. The datasets were collected at 1000 Hz of sampling rate. However, for analysis purposes, they were down-sampled to 100 Hz. The description of the dataset is available in [21].

C3, and C4 channels, which are over the sensorimotor area, are known to record significant motor imagery characteristics [22]. From the study of [13], the Fp2 channel, which is in the forehead, was found to have comparable high classification accuracies with C4. Besides that, the AF3 channel which is also located near the forehead was ranked as the most informative channel in the previous study [23]. As a result, the EEG signal from C3, C4, Fp1, Fp2, AF3, and AF4 channels are chosen among 118 channels for further processing and analysis.

B. Pre-processing

The pre-processing step has been used to further remove any artifacts from the signal. In this study, the pre-processing phase consists of three steps. The 5th-order Butterworth filter with a pass band of 8- 30 Hz was used in the first step as suggested in [23], [24]. Therefore, the raw EEG signal was passed through the filter to extract two EEG frequency sub-bands (α and β rhythms) and remove all the remaining ranges. Most signals below 5 Hz have artifacts [19]. The power-line interference at 50 Hz was removed from the EEG signals using a notch filter in the second step. Lastly, the signals were segmented to extract the dataset's epochs with three different window lengths.

C. Features Extraction

In Experiment 1 (E1), the discrete wavelet transform technique (DWT) was used to decompose the signal. It is one of the wavelets transform (WT) techniques. It is a decomposition technique providing both time and frequency information. It divides the signal into low-frequency components (approximation coefficients) and high-frequency components (detail coefficients) using the low-pass filter (LPF) and high-pass filters (HPF) respectively. The sample then is down sampled by 2. Based on the required level, the approximate components are further decomposed for multi-stage decomposition depending on the tasks to be performed [25]. In this study, signals were decomposed using three levels of DWT with Daubechies 4 wavelet (db4) producing three details components (D1, D2 and D3), and one approximation component (A3). Since the frequency band of A3 (0-6.25 Hz) is less than 5 Hz, it may have artifacts [19]. Therefore, five features were extracted from each detail component producing a total of fifteen (15) features in E1. The detail components are shown in Fig.2: D1 (25-50 Hz), D2 (12.5- 25 Hz), and D3 (6.25-12.5 Hz). The following statistical features were extracted in E1 for classification:

The absolute mean (μ) is written as [26]:

$$\mu = \frac{1}{N} \sum_{n=1}^N |y_n| \quad (1)$$

The standard deviation (σ) is written as [20]:

$$\sigma = \sqrt{\frac{1}{N} \sum_{n=1}^N (y_n - \bar{y})^2} \quad (2)$$

The skewness is defined as [20]:

$$skewness = \frac{\frac{1}{N} \sum_{n=1}^N (y_n - \bar{y})^3}{\sigma^3} \quad (3)$$

The kurtosis is defined as [20]:

$$kurtosis = \frac{\frac{1}{N} \sum_{n=1}^N (y_n - \bar{y})^4}{\sigma^4} \quad (4)$$

The average power of coefficients (P_{av}) is written as [26]:

$$P_{av} = \sqrt{\frac{1}{N} \sum_{n=1}^N y_n^2} \quad (5)$$

where N is the number of samples, y_n is the signal in each sub-band and $\bar{y}$ is a mean of the signal in each sub-band. n is an integer that belongs to 1 to N .

In Experiment 2 (E2), the mean absolute value, Root Mean Square (RMS), Hjorth parameters (activity, mobility, complexity), and waveform duration were six features that were extracted in the time domain. They were extracted directly from the processed signal. The absolute mean value (μ) is written as [26] :

$$\mu = \frac{1}{T} \sum_{t=1}^T |x_t| \quad (6)$$

RMS is the square root of the average of the signal's squared value in the time domain. The RMS is written as [27]:

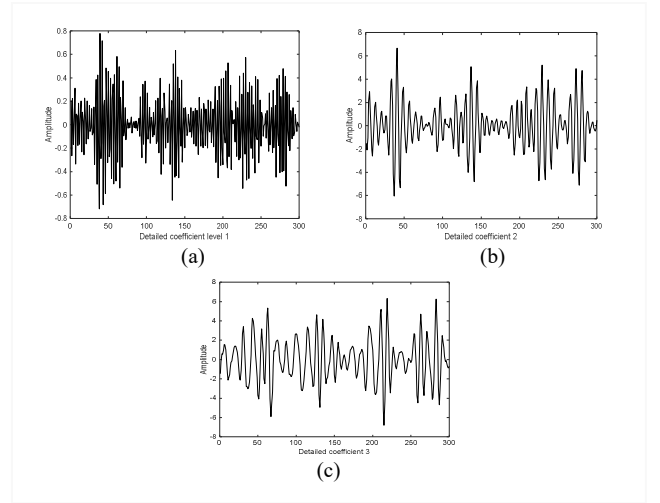


Fig. 2. Wavelet decomposition sub-band signals of (a) detailed at level 1 (D_1), (b) detailed at level 2 (D_2) and (c) detailed at level 3 (D_3).

$$RMS = \sqrt{\frac{1}{T} \sum_{t=1}^T x_t^2} \quad (7)$$

The activity is measured using the amplitude variance of signal samples [28], [29]. It is defined as:

$$Activity = var(x_t) \quad (8)$$

The mobility provides the mean approximation of frequency [28], [29]. It is defined as:

$$Mobility = \sqrt{\frac{var(x'_t)}{var(x_t)}} \quad (9)$$

The complexity gives the standard deviation of the power spectrum [28], [29]. It is defined as:

$$Complexity = \frac{Mobility(x'_t)}{Mobility(x_t)} \quad (10)$$

Waveform length (WL) represents the cumulative absolute difference between adjacent samples and provides a measure of the signal's overall variation. It can be expressed as follows [30]:

$$WL = \sum_{t=1}^T |x_t - x_{t-1}| \quad (11)$$

where T is the number of samples and x_t is the processed signal. x'_t is the first derivative of the signal sample x_t . $var(x_t)$ is the variance of the signal sample x_t . t is an integer that belongs to 1 to T .

In Experiment 3 (E3), the feature sets in E1 and E2 were combined. Their effects on the performance were evaluated.

D. Classification

Two classifiers, Support Vector Machine (SVM) and Logistic Regression (LR) were employed to evaluate the performance of all feature sets in three experiments. SVM has good generalization and has been effectively used to a relatively large number of BCI while LR is relatively simple algorithm to implement and interpret even with small sample sizes [31]– [33]. The features were normalized before the

evaluation. The accuracy of the entire data was used to determine the classifier's capacity to perform successfully classifying the motor tasks. The definition of the metric is:

$$Accuracy = \frac{\text{Number of correct predictions}}{\text{Number of all predictions}} \times 100 \quad (12)$$

The accuracy of the classifiers was assessed using 10-fold cross-validation. The data is divided into ten equal-sized subsets or folds through random splitting. It comprises nine of the folds is a training set and another fold is the testing set. In each round, the model is trained on nine of the folds and evaluates the performance on the remaining fold. It is repeated ten times and produces ten classification accuracy. The average of the classification accuracy is computed. The classification was implemented on MATLAB R2022a.

IV. RESULTS AND DISCUSSION

A. Time Range Selection

Table I and Table II show classification accuracy (%) for different time ranges and channels when the features vector of all participants is combined. The features set is only from E3 combining time-frequency and time-domain features. The data samples were divided into four sets of time windows: 3.5-second, 3.0-second, and two sets of 2-second time windows with different EEG channels. The time ranges are 0 s to 3.5 s, 0.5 s to 3.5 s, 0.5 s to 2.5 s, and 1 s to 3 s as shown in Table I and Table II. In order to identify the significance of the specified time windows and channel selection on the classification accuracy, a two-way analysis of variance (ANOVA) was employed as suggested in the previous study [13].

The result shows that there are no significant differences in time range selection for either classifier; SVM (0.32451 > p=0.05) or LR (0.61542 > p=0.05). Since the mean accuracy in the time range of 0.5 s to 3 s is slightly higher than other time ranges for both classifiers, further analysis is based on the data sample set with the range of 0.5 s to 3 s. The best accuracy would vary in a different time range for each participant [13]. The benefit of reducing the window length can increase the Information Transfer Rate (ITR) [34]. However, the metric is not used in this paper to evaluate the classification performance.

TABLE I. ACCURACY VALUES FOR DIFFERENT TIME RANGES (SVM)

Channel	Time Range			
	0-3.5s	0.5-3.5s	0.5-2.5s	1-3s
Fp1	63.0	64.2	62.6	62.4
Fp2	61.0	61.2	59.5	62.1
AF3	58.9	61.1	64.6	60.1
AF4	59.9	61.1	59.1	60.4
C3	66.9	67.7	64.7	68.6
C4	61.6	64.1	62.0	64.6
Mean±SD	61.9±2.8	63.2±2.6	62.1± 2.4	63.0±3.2

TABLE II. ACCURACY VALUES FOR DIFFERENT TIME RANGES (LR)

Channel	Time Range			
	0-3.5s	0.5-3.5s	0.5-2.5s	1-3s
Fp1	63.9	65.4	63.1	63.6
Fp2	61.4	61.9	59.5	62.9
AF3	58.5	60.5	66.1	59.8
AF4	59.5	61.1	57.8	60.5
C3	66.2	67.3	65.5	67.6
C4	61.7	62.6	62.3	63.1
Mean±SD	61.9±2.8	63.1±2.7	62.4±3.3	62.9±2.8

There are significant differences in channel selection for either classifiers; SVM (0.00012 < p-value=0.05) or LR (0.00096 < p-value=0.05). Even though the C3 channel has the highest average accuracy, all the selected channels are included in the single-channel EEG signal analysis in the next subsection.

B. Classification Performance of SVM

Due to inter-subject variation of MI task responses, the performance of the model is different among participants. Therefore, Table III provides the classification accuracy of each participant that uses SVM as the classifier. The greatest classification accuracy of the channel is presented in boldface, while the lowest classification accuracy is shown in italics and boldface.

'**al**' (C3) achieves the best classification accuracy of 87.1% with the features in E3. The channel is widely used for analysis in previous works since it is located over the sensorimotor area and can record important characteristics [17], [19], [26]. Besides that, all channels at the forehead achieve the classification accuracy that is greater than 70% from '**al**'. The result is supported by a previous study, which found that EEG channels on the forehead could identify motor imagery just as well as common channels [13]. Users are considered BCI illiterate if they do not reach a level of accuracy greater than 70% [35]. For AF4, the accuracy of '**al**' is increased to 70.4% with the features in E3 which is 13.6% higher than E1 and 5% higher than E2. The lowest classification accuracy is 46.4 % that obtained by '**aa**' (Fp1) when implementing E1. In general, when a features matrix based on DWT is applied, E1 has the lowest classification accuracy. Certain participants may produce unstable signals for MI tasks. Consequently, the algorithm may fail to decode the tasks with high accuracy. Besides that, different types of wavelet transform, and decompositions may fail to extract non-linear relationships inside a signal even though they are suited for biomedical signal analysis [26].

C. Classification Performance of LR

Table IV provides the performance of each participant that uses LR as the classifier. The highest accuracy is also achieved by '**al**' (C3). The accuracy is 82.5% with the features in E1 and E2. Meanwhile, the lowest classification accuracy is 45.4% which is also from '**aa**' (Fp1). However, '**al**' have an accuracy of more than 70% on all channels in the forehead

area. For AF4, the accuracy is again increased up to 70% by using the features in E3 which is 14.6% higher than E1 and 5.7% higher than E2. Combining time-frequency features with time-domain features has increased the accuracy and robustness of the MI tasks on certain channels as well as participants. When LR is used as the classifier, the features in E1 also contribute to the lowest classification accuracy.

TABLE III. ACCURACY VALUES IN DIFFERENT EXPERIMENTS (SVM)

CHN	EXP	Participants					
		<i>aa</i>	<i>al</i>	<i>av</i>	<i>aw</i>	<i>ay</i>	<i>all</i>
Fp1	E1	46.4	61.4	54.6	52.5	49.3	54.0
	E2	57.1	73.9	55.7	63.2	66.1	65.2
	E3	60.4	73.2	50.7	61.8	64.3	64.2
Fp2	E1	53.6	48.2	53.9	51.8	47.9	53.2
	E2	56.4	74.6	53.6	68.2	65	62.4
	E3	54.3	72.9	58.6	62.9	59.6	61.2
AF3	E1	47.1	58.6	53.9	58.9	54.6	58.1
	E2	51.1	71.8	56.8	57.5	65	59.6
	E3	48.2	67.1	58.2	59.6	62.5	61.1
AF4	E1	46.5	56.8	55	53.2	51.4	56.0
	E2	56.8	65.4	52.9	60	63.6	58.7
	E3	52.9	70.4	58.2	58.9	57.9	61.1
C3	E1	63.2	83.9	57.5	53.9	75.7	66.1
	E2	63.2	84.3	57.5	60.7	76.4	67.7
	E3	63.2	87.1	58.6	61.4	76.8	67.7
C4	E1	52.9	73.6	59.3	77.5	52.5	63.1
	E2	59.6	69.6	61.4	77.9	54.3	63.1
	E3	52.9	77.5	61.8	78.2	53.6	64.1

TABLE IV. ACCURACY VALUES IN DIFFERENT EXPERIMENTS (LR)

CHN	EXP	Participants					
		<i>aa</i>	<i>al</i>	<i>av</i>	<i>aw</i>	<i>ay</i>	<i>all</i>
Fp1	E1	45.4	59.6	54.6	50.7	52.1	54.1
	E2	56.1	75.7	56.1	63.6	64.6	65.4
	E3	66.8	75	50.4	65	67.5	65.4
Fp2	E1	55.4	53.9	49.6	49.6	50.4	53.4
	E2	57.1	73.9	54.3	68.2	65.7	62.1
	E3	57.1	72.9	55.4	65.4	60.7	61.9
AF3	E1	53.2	57.1	52.9	56.1	61.1	57.4
	E2	52.1	70.7	56.8	56.1	66.1	59.3
	E3	54.6	72.9	53.2	61.8	63.6	60.5
AF4	E1	52.9	55.4	54.3	56.8	53.6	56.5
	E2	56.1	64.3	54.6	63.2	62.1	58.5
	E3	59.3	70	53.9	63.6	62.9	61.1
C3	E1	57.5	82.5	55	55.7	74.3	66.8
	E2	61.8	82.5	56.8	61.4	75.4	68.1
	E3	58.9	81.8	57.1	60.4	75.4	67.3
C4	E1	54.1	72.9	54.6	77.9	51.8	63.9
	E2	58.9	70.4	58.9	78.6	51.1	61.8
	E3	50.0	72.9	59.3	77.9	50.0	62.6

V. CONCLUSION

The feature extraction and classification strategies for MI-based EEG data have been discussed for the EEG single-channel implementation in this paper. The time range of 0.5 to 3.5 seconds was chosen based on the average of the classification accuracy, even though there were no significant differences between the accuracy of the time ranges for both classifiers. By considering the results for each participant, it was found that with the features in E3, the accuracy of "*al*" through the AF4 channel was increased up to 70% and 70.4% by using SVM and LR respectively. This accuracy was higher than in E1 and E2. When SVM was used as the classifier, '*al*' through the C3 channel delivered the highest classification accuracy of 87.1% in E3. The combination of time-frequency domain and time-domain features as implemented in E3 is one of the approaches to overcome the source limitation of single-channel EEG signal implementation. As a result, the proposed method has the potential for executing the single-channel EEG with greater detection of MI tasks particularly at the area of the forehead to enhance the practicality as stated in previous work.

VI. REFERENCES

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